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Paper 1: A Systematic Review and Model-Based Meta-Analysis of Pegylated-Interferon- α -Induced

Bibliographic Information

- **Title:** A Systematic Review and Model-Based Meta-Analysis of Pegylated-Interferon- α -Induced HBsAg Loss in Chronic Hepatitis B Virus Infection
- **Authors:** N.J.H. et al.
- **Journal:** CPT: Pharmacometrics & Systems Pharmacology, 2025
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Paper Type: POPK_PKPD

Quick Take

This MBMA quantifies HBsAg loss rates with Peg-IFN α -based regimens in chronic HBV, identifying baseline HBsAg, treatment duration, and HBeAg status as key predictors. It provides a quantitative framework for clinical trial design and enrichment strategies. Pharmacometricians involved in HBV drug development or model-informed drug development will find this useful for benchmarking and simulation.

Executive Summary

This first model-based meta-analysis of Peg-IFN α -induced HBsAg loss integrates data from 45 studies (EOT) and 28 studies (24-week post-treatment). Baseline HBsAg and treatment duration significantly predicted response in both models, with HBeAg status additionally significant at EOT. Covariates reduced between-trial variance by 58-78%, enabling more precise predictions. The model supports trial simulation, sample size estimation, and has been used to calibrate in silico trials, demonstrating its utility in advancing model-informed drug development for chronic hepatitis B.

Scientific Context & Motivation

HBsAg loss is the primary goal of CHB treatment, but response to Peg-IFN α varies widely across populations and regimens. Traditional meta-analyses fail to account for between-trial heterogeneity or identify predictors. This MBMA addresses the gap by quantifying variability and identifying key covariates to inform trial design and personalized treatment strategies.

✂ Methodological Snapshot

A model-based meta-analysis using logistic regression with random effects on the logit scale. Two cross-sectional models were developed for HBsAg loss at EOT and 24 weeks post-treatment. Covariates were tested stepwise, with baseline HBsAg, treatment duration, and HBeAg status

retained. Between-trial variance was modeled as a random intercept. Simulations incorporated parameter uncertainty and between-trial variability.

☒ Structural Model Breakdown

The model is a logistic regression with random intercepts: $\text{logit}(p) = \beta_0 + \beta_1 * \text{Drug} + \beta_2 * \text{Duration} + \beta_3 * \text{HBsAg} + \beta_4 * \text{HBeAg} + \eta$, where $\eta \sim N(0, \omega^2)$. Drug effect is categorical (Peg-IFN α vs. Peg-IFN α +NA). Duration centered at 48 weeks, HBsAg centered at 3 log10 IU/mL, HBeAg binary (positive vs. negative). Between-trial variance ω^2 estimated as 0.484 (EOT) and 0.572 (24-week). The model does not include interaction terms or non-linear effects.

Detailed Methodological Analysis

Modeling Approach

Logistic regression with random intercepts (logit scale) using nlme in R. Base model estimated overall probability; treatment-effect model distinguished Peg-IFN α monotherapy vs. combination with NA; final covariate model included significant predictors. Covariates centered to median values for interpretability.

Data Sources

Systematic review of Embase, MEDLINE, Cochrane (Jan 2000-Jul 2022). Included RCTs, prospective/retrospective cohorts, single-arm trials. 45 studies (83 arms, 11,493 patients) for EOT; 28 studies (58 arms, 4267 patients) for 24-week. ITT/mITT prioritized; per-protocol excluded. Study-strata-arms defined by treatment arm and stratification factors (HBeAg, baseline HBsAg).

Estimation Methods

Restricted maximum likelihood (REML) via nlme package in R. Covariate selection using $p < 0.05$ in stepwise forward selection.

Model Evaluation

AIC, goodness-of-fit plots (observed vs. individual-predicted, weighted residuals vs. population predictions and baseline HBsAg), parameter precision (%RSE). Prediction intervals calculated from fixed-effect variance-covariance matrix and between-trial variance.

Covariate Analysis

Stepwise forward selection. Tested: baseline HBsAg (log10 IU/mL), Peg-IFN α treatment duration (weeks), baseline HBeAg status (positive vs. negative), age, gender, race (proportion non-Asian), study design (RCT vs. non-RCT), continuation of NA after Peg-IFN α (24-week model only). Significant: baseline HBsAg ($p < 0.001$), treatment duration ($p < 0.01$), HBeAg status (EOT only, $p = 0.007$).

Statistical Rigor Assessment

Appropriate use of random-effects meta-analysis to account for between-trial heterogeneity. Covariate selection based on p-values with no multiplicity correction, which may inflate Type I error. Sample size is large (83 and 58 arms). Missing data handled by excluding studies with incomplete covariate information. Sensitivity analyses not reported. Prediction intervals appropriately reflect both parameter uncertainty and between-trial variability. The use of study-level aggregates may lead to ecological fallacy, but this is inherent to MBMA.

Key Findings

Baseline HBsAg was the strongest predictor: each 1 log₁₀ IU/mL increase reduced odds of HBsAg loss by 79% in both models. Treatment duration had a modest effect (1.3% increase in odds per week at EOT, 3.1% at 24 weeks). HBeAg-positive patients had 2.37-fold higher odds at EOT. In the reference population (HBeAg-negative, baseline HBsAg 3 log₁₀, 48 weeks), Peg-IFN α monotherapy yielded 8.5% HBsAg loss at EOT and 11.1% at 24 weeks; combination with NA gave 11.7% and 14.1% respectively. Simulations showed that selecting patients with lower baseline HBsAg (e.g., ≤ 2 log₁₀) could increase loss rates to 34-48%.

Clinical & Regulatory Implications

The model supports patient enrichment in clinical trials by selecting those with low baseline HBsAg and HBeAg-positive status to increase HBsAg loss rates. Treatment duration of 48 weeks is standard; longer duration may modestly improve odds. The MBMA provides a benchmark for control arm response rates, enabling sample size estimation and go/no-go decisions. It has been used to calibrate in silico trials, demonstrating regulatory utility. The findings can inform combination therapy strategies where Peg-IFN α is added to novel agents.

Strengths & Limitations

Strengths

- First MBMA for HBsAg loss with Peg-IFN α , providing a quantitative framework
- Large dataset (45 studies, 11,493 patients for EOT; 28 studies, 4267 for 24-week)
- Identified key predictors (baseline HBsAg, treatment duration, HBeAg) that substantially reduced between-trial variance
- Prioritized ITT/mITT results, reducing bias from per-protocol analyses
- Applied to calibrate and validate in silico trials, demonstrating translational utility

Limitations (Acknowledged by Authors)

- Reliance on aggregated published data, limiting ability to explore individual-level covariates
- Missing data for HBV genotype, baseline HBV DNA, and ALT, which are known predictors
- Predominance of Asian populations, limiting generalizability to other ethnicities
- Residual between-trial heterogeneity remained after covariate inclusion

Limitations (Expert Review)

- Use of study-level means/medians may obscure individual-level relationships and introduce ecological bias
- Exclusion of per-protocol results may introduce selection bias if ITT data were not available for all studies
- No assessment of publication bias or small-study effects
- Assumption of linear covariate effects on logit scale may not capture non-linear relationships
- No sensitivity analyses for missing data or model assumptions

Generalizability

Findings are generalizable to CHB patients treated with Peg-IFN α , but predominantly Asian populations limit extrapolation to other ethnicities. The models are based on aggregated data and may not capture individual-level heterogeneity fully. The results are most applicable to clinical trial design and enrichment strategies rather than individual patient predictions.

Figures & Tables

- **Table 1:** Study characteristics for EOT and 24-week post-treatment models, including number of studies, study-strata-arms, participants, and treatment distribution by study design.
 - *Significance:* Provides an overview of the data used in the MBMA, showing the breadth of evidence and balance across treatment arms and study designs.
- **Table 2:** Parameter estimates for final covariate models at EOT and 24 weeks post-treatment, including logit coefficients, transformed probabilities/odds ratios, %RSE, 95% prediction intervals, and p-values.
 - *Significance:* Core results table showing the magnitude and significance of covariate effects, enabling readers to understand the model structure and apply the findings.
- **Table 3:** Simulated HBsAg loss rates for hypothetical phase 3 trial populations stratified by baseline HBsAg criteria and HBeAg status, for both Peg-IFN α monotherapy and combination therapy.
 - *Significance:* Demonstrates the practical utility of the MBMA for trial design, showing how patient selection affects predicted response rates.
- **Figure 1:** Forest plot comparing estimated HBsAg loss rates from base and final covariate models at EOT and 24 weeks post-treatment, with prediction intervals.
 - *Significance:* Visually illustrates the reduction in between-trial variability and narrowing of prediction intervals after covariate inclusion.
- **Figure 2:** Goodness-of-fit plots for the EOT model: observed vs. individual-predicted HBsAg loss and weighted residuals vs. population predictions and baseline HBsAg.
 - *Significance:* Validates the EOT model, showing no systematic bias and adequate fit across the range of observed responses.
- **Figure 3:** Goodness-of-fit plots for the 24-week post-treatment model: observed vs. individual-predicted HBsAg loss and weighted residuals vs. population predictions and baseline HBsAg.
 - *Significance:* Validates the 24-week model, confirming adequate fit and no residual trends.

Future Directions

Inclusion of individual patient data to explore additional covariates (genotype, ALT, HBV DNA). Longitudinal modeling of HBsAg decline over time. Application to novel combination therapies (e.g., with siRNA or ASO). Validation in prospective trials. Extension to other endpoints such as HBeAg seroconversion or HBV DNA suppression.

Expert Commentary

This MBMA is a well-conducted analysis that addresses a key gap in HBV functional cure research. The identification of baseline HBsAg as the dominant predictor is consistent with clinical experience and provides a practical tool for trial enrichment. The reduction in between-trial variance is impressive, though residual heterogeneity suggests unmeasured factors remain. The use of aggregated data limits the ability to explore non-linear or time-varying effects. The integration with in silico trials is a forward-looking application that showcases the value of MBMA in model-informed drug development. Future work should focus on individual-level data and dynamic modeling of HBsAg kinetics.

Bottom Line

This MBMA provides a validated quantitative framework for predicting HBsAg loss with Peg-IFN α , identifying baseline HBsAg as the dominant predictor. It enables evidence-based trial design and enrichment strategies, and has been successfully used to calibrate mechanistic in silico models. Pharmacometricians should consider these results when designing HBV trials or evaluating new combination therapies.

☒ Figures

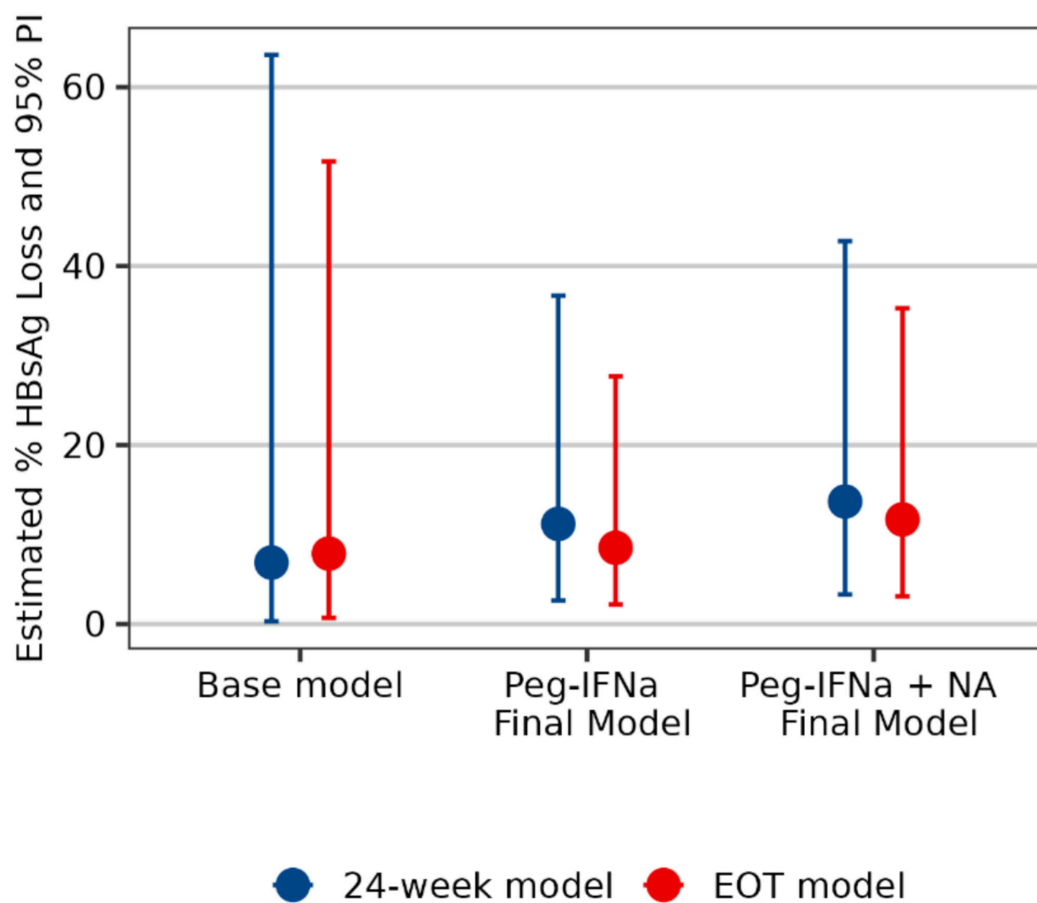


Figure 1: Estimated hepatitis B surface antigen (HBsAg) loss at end of treatment (EOT) and 24 weeks post-treatment from the base and final covariate models, using data fro

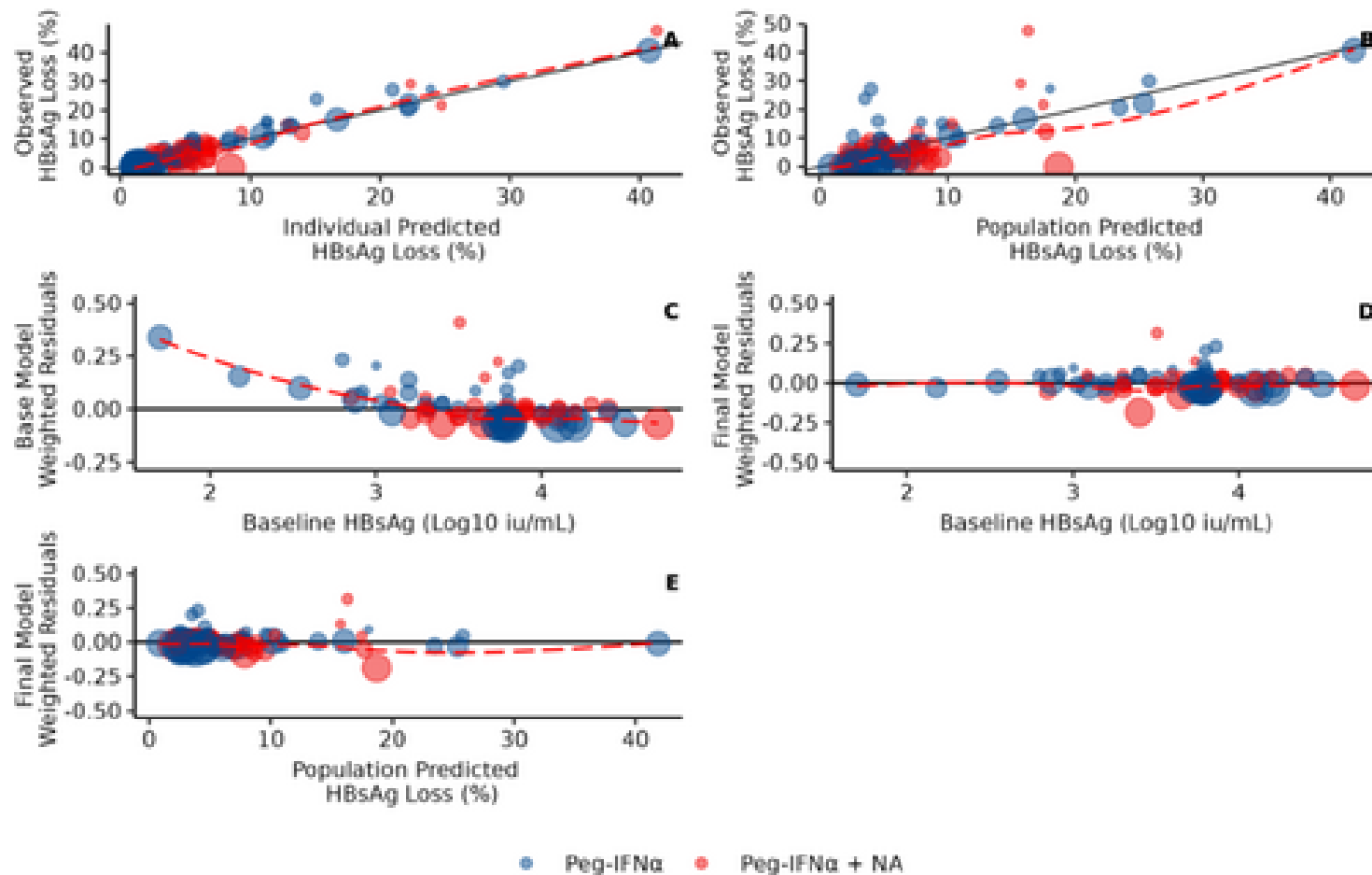


Figure 2: Goodness-of-fit for the hepatitis B surface antigen (HBsAg) loss model at end of treatment (EOT). (A) Observed versus individual-predicted HBsAg loss with the li

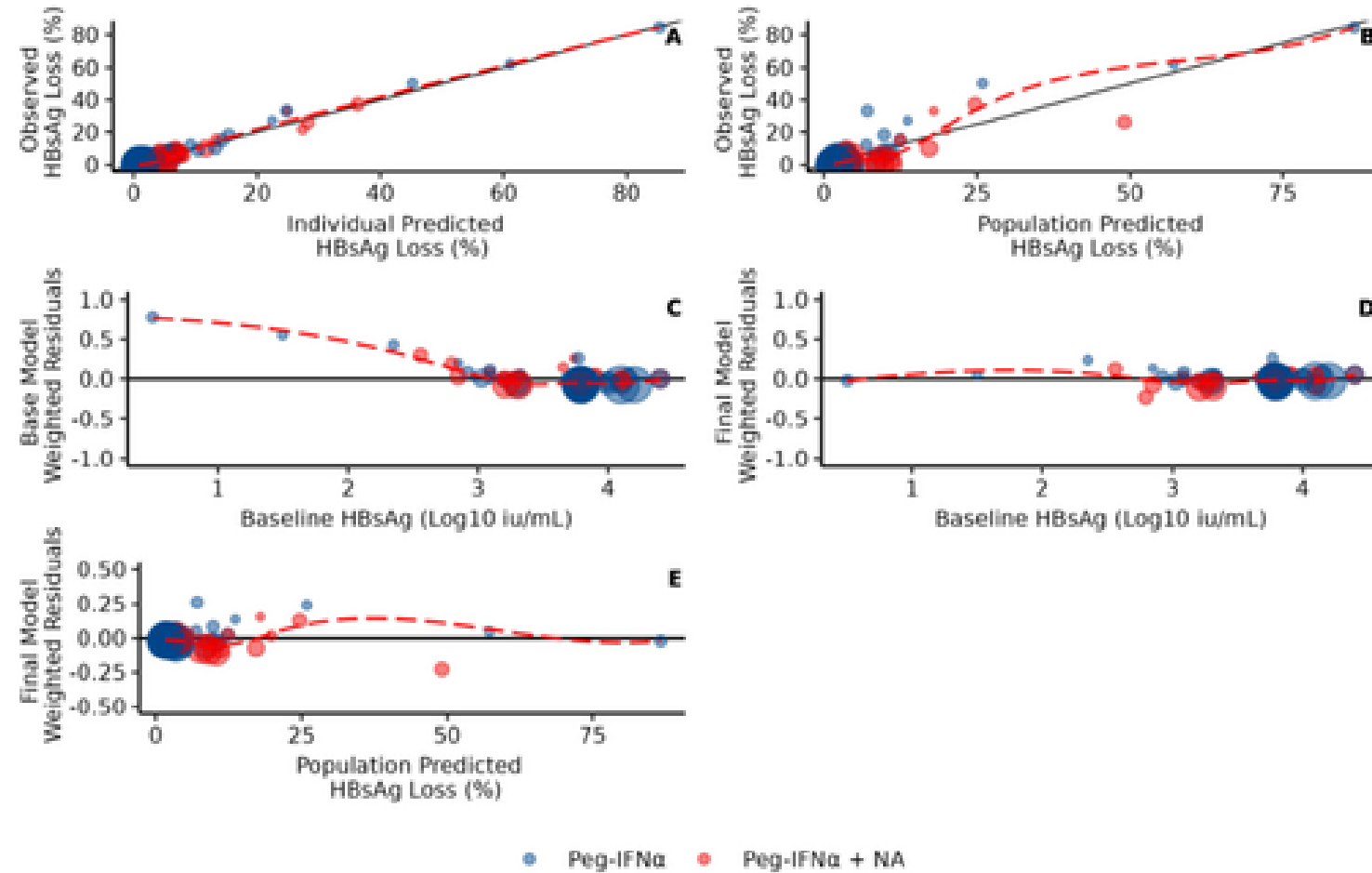


Figure 3: Goodness-of-fit for the hepatitis B surface antigen (HBsAg) loss model at 24 weeks post-treatment. (A) Observed versus individual-predicted HBsAg loss with the 1